MACHINE LEARNING IN PHYSICS
AUTOENCODERS

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PHY6937 / PHY4936

Introduction (1)

- An autoencoder approximates the identity map

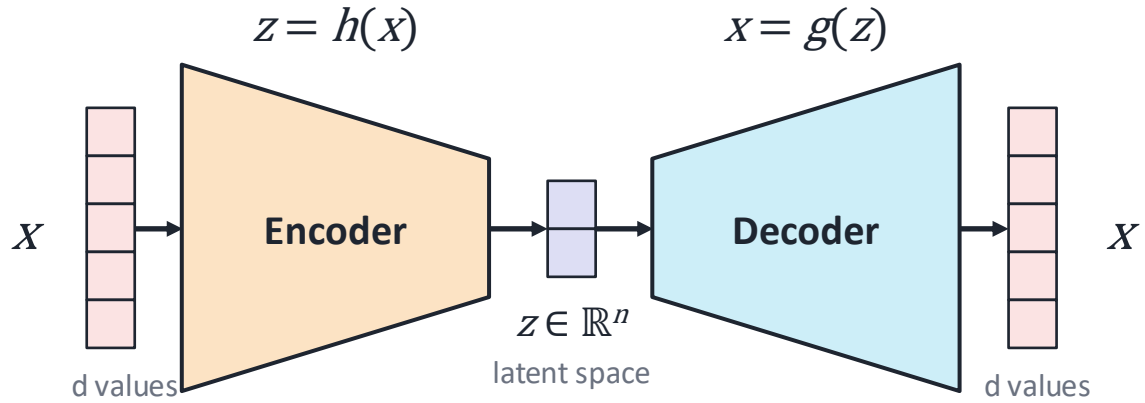
$$f: x \to x, \quad x \in \mathbb{R}^d$$

- But there is an important twist: the map f is a composition $f(x) = (g \circ h)(x)$ where
  $h: x \to z, \ z \in \mathbb{R}^n$ followed by $g: z \to x$
  that is, $z = h(x)$ followed by $x = g(z)$

- The function $z = h(x)$ is called the encoder and $x = g(z)$ is called the decoder.

Introduction (2)

- The space $\mathbb{R}^n$ is called the latent space.
- Typically, $n \ll d$.
- The encoder *compresses* the input data x



in an *unsupervised* manner, that is, the *labels* are the data x.

Introduction (3)

Autoencoders have multiple applications including:

1. Data compression
    1. Lossy compression
    2. Structure detection
2. Anomaly detection
3. Noise removal

Our example focuses on structure detection.

Introduction (4)

The key feature of an autoencoder is the bottleneck between the encoder and the decoder. This forces the model to construct a lower-dimensional representation in the latent space $z \in \mathbb{R}^n$ of the input data $x \in \mathbb{R}^d$.

Since the goal is to model the mapping $f: x \to x$, the *quadratic* loss, $L(x, f) = (x - f)^2$, is typically used. Revisiting our derivation of the functional derivative $\delta R / \delta f$, but specializing to the current case, leads to

$$\int dx\, p(x) g(x) \frac{\partial L}{\partial f} = 0$$

where L is directly the loss rather than a functional of it.

Introduction (5)

$$\int dx\, p(x) g(x) \frac{\partial L}{\partial f} = 0$$

Since, by assumption, the function $g(x)$ is arbitrary (and assuming $p(x) > 0 \; \forall \; x$) the above will hold *if and only if*

$$\frac{\partial L}{\partial f} = 0$$

which, for $L(x, f) = (x - f)^2$, implies

$$f_{\omega^*}(x) = x.$$

In principle, the autoencoder provides an *unbiased* estimate of the input data.

Introduction (6)

- There are many kinds of autoencoders, but we'll consider two of the simplest:
  - (Standard) Autoencoder (AE)
  - Convolutional Autoencoder (CAE)

- An important class of autoencoders that we shall not discuss is Variational Autoencoders (VAE).

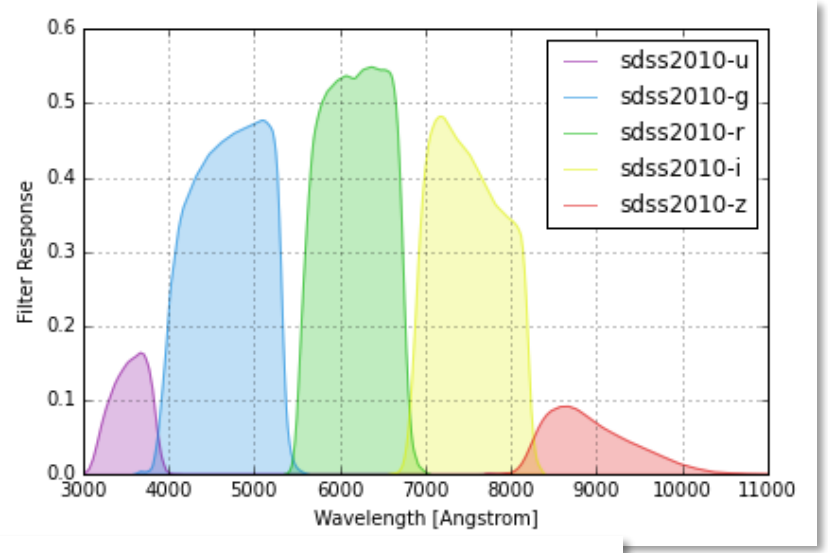
- We'll discuss AE and CAE using simple examples.

AUTOENCODERS (AE)

Autoencoders (1)

- The autoencoder will be used to search for structure in stellar color data from the Sloan Digital Sky Survey (SDSS)*.

- The data are fluxes measured in five filters:



Filter	Wavelength (nm)
Ultraviolet (u)	354.3
Green (g)	477.0
Red (r)	623.1
Near Infrared (i)	762.5
Infrared (z)	913.4

*https://www.sdss.org/

Autoencoders (2)

SDSS Dataset (https://skyserver.sdss.org/dr20/SearchTools/sql)

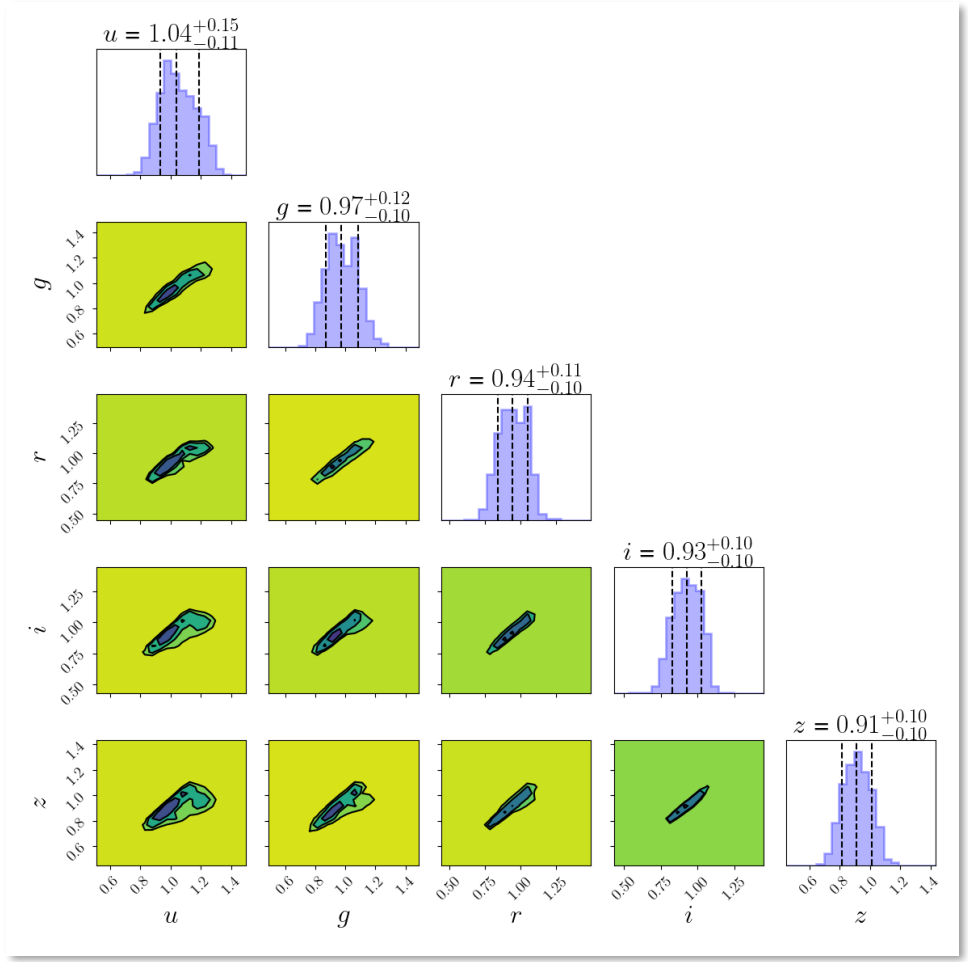
We use data from 255,000 stars extracted using the SQL command:

```
SELECT TOP 255000
  p.ra, p.dec, s.z as redshift, p.u, p.g, p.r, p.i, p.z
FROM PhotoObj AS p
  JOIN SpecObj AS s ON s.bestobjid = p.objid
WHERE
       p.u BETWEEN 0 AND 30
  AND p.g BETWEEN 0 AND 30
  AND p.r BETWEEN 0 AND 30
  AND p.i BETWEEN 0 AND 30
  AND p.z BETWEEN 0 AND 30
  AND s.class = 'STAR'
```

Autoencoders (3)

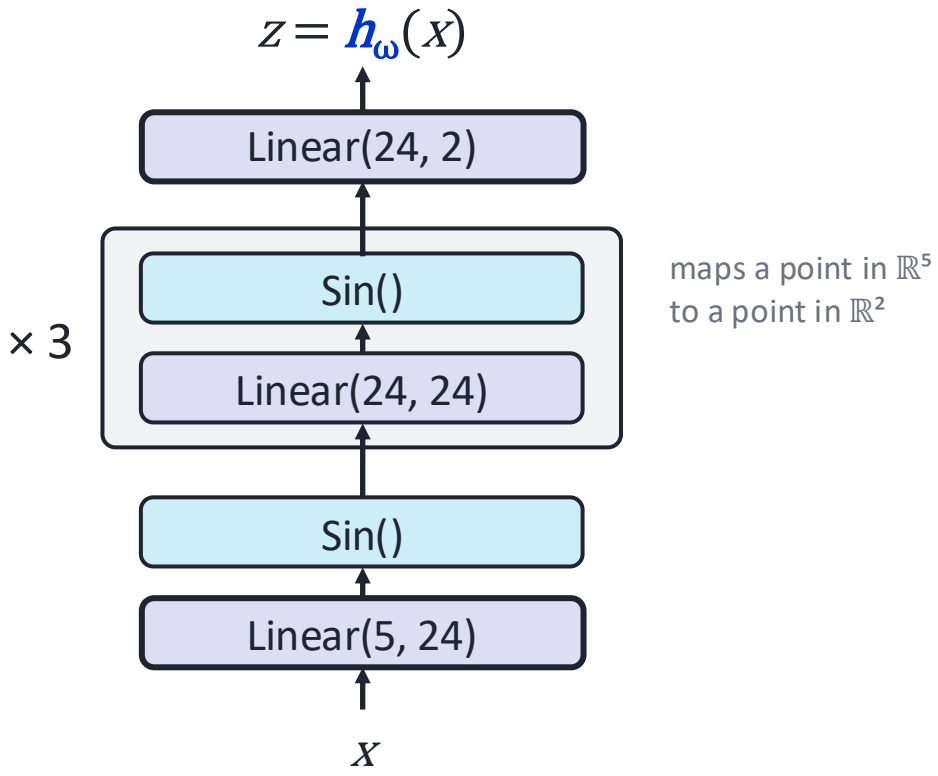
Beyond correlations, the plots show no obvious structure.

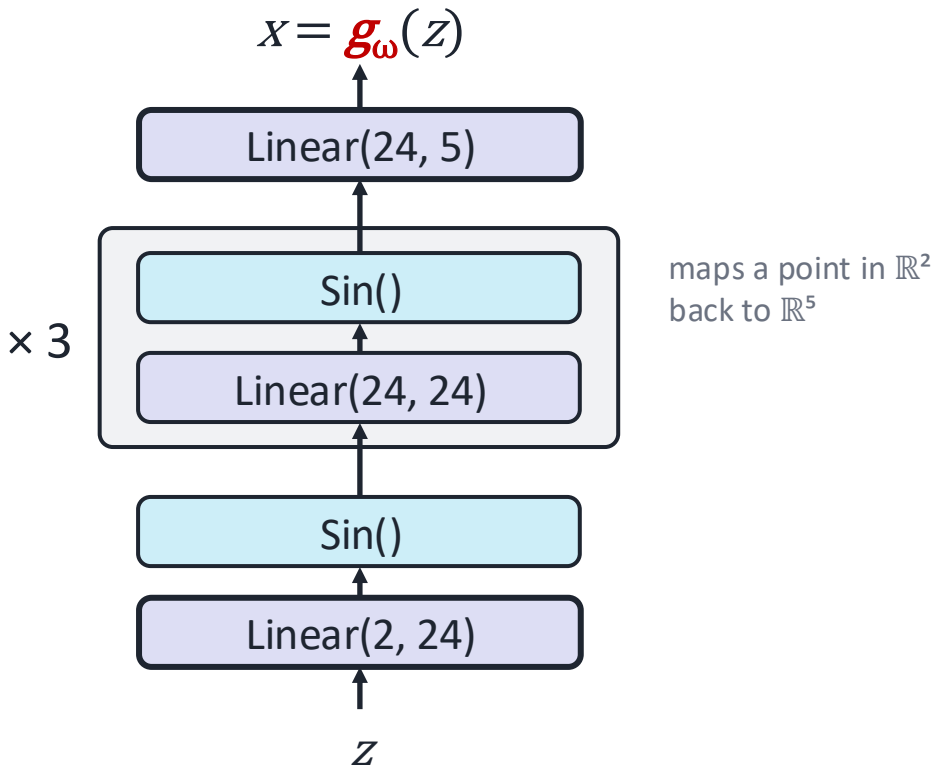
But the hope is that the latent space of the autoencoder will reveal any structure that may be present in the data.



Autoencoders (4): Encoder / Decoder

$z \in \mathbb{R}^2, \qquad x \in \mathbb{R}^5$





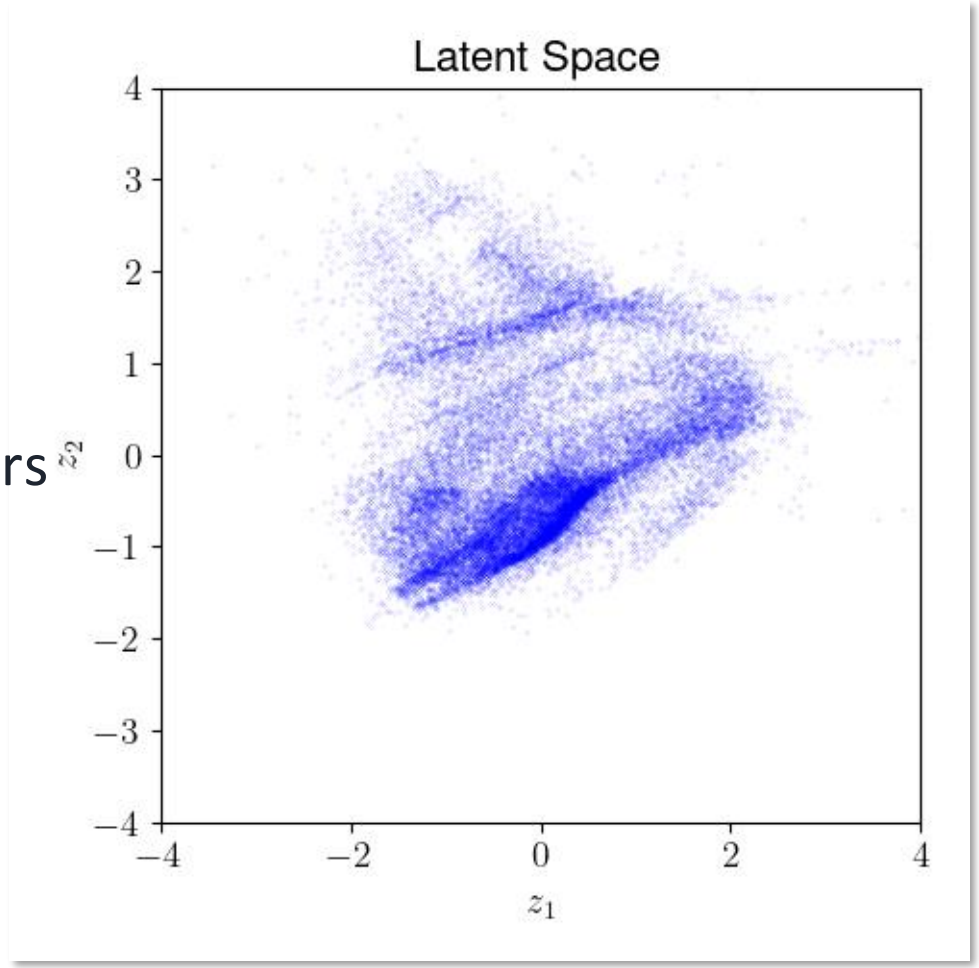
Autoencoders (5): Training

Configuration: AE_config.yaml

```
train_size:       150_000
batch_size:       128
n_epochs:         1000
test_size:        100_000
val_size:         5000
n_steps:          4
n_iters_per_step: 293_000
n_iterations:     1_172_000
base_lr:          0.002
gamma:            0.8
```

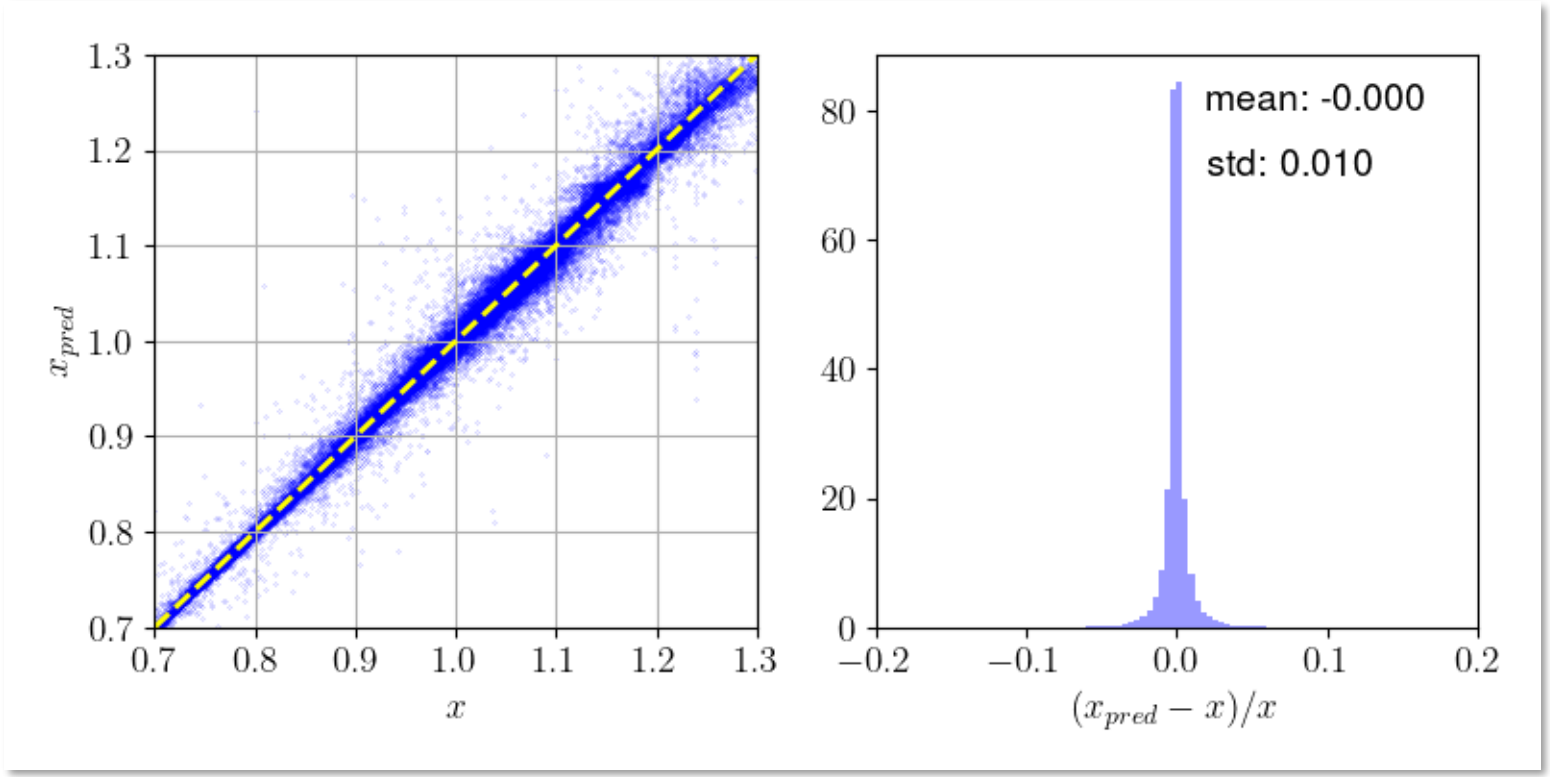
Autoencoders (6): Latent Space

- There is clearly structure in the latent space, suggesting (as expected) that the color data pertains to different classes of stars.
- The next step would be to identify the clusters and determine whether they correspond to stars with physically distinct characteristics.



Autoencoders (7): Predictions

- In the plots below x is the measured flux and x_{pred} the predicted flux. As anticipated, the bias in the autoencoder is small.



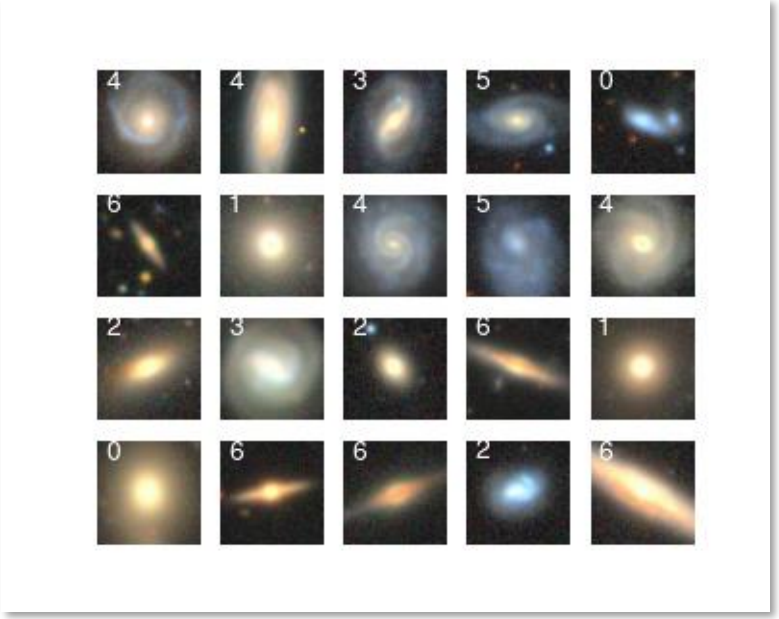
CONVOLUTIONAL AUTOENCODERS (CAE)

Convolutional Autoencoders (1)

- A **C**onvolutional **A**uto**E**ncoder is an autoencoder that operates on 2-dimensional data, such as images.

- Suppose our goal is to create a compressed representation of galaxy images.

- This can be done using a CAE with an encoder and a decoder which, conceptually, are mirror images of each other.

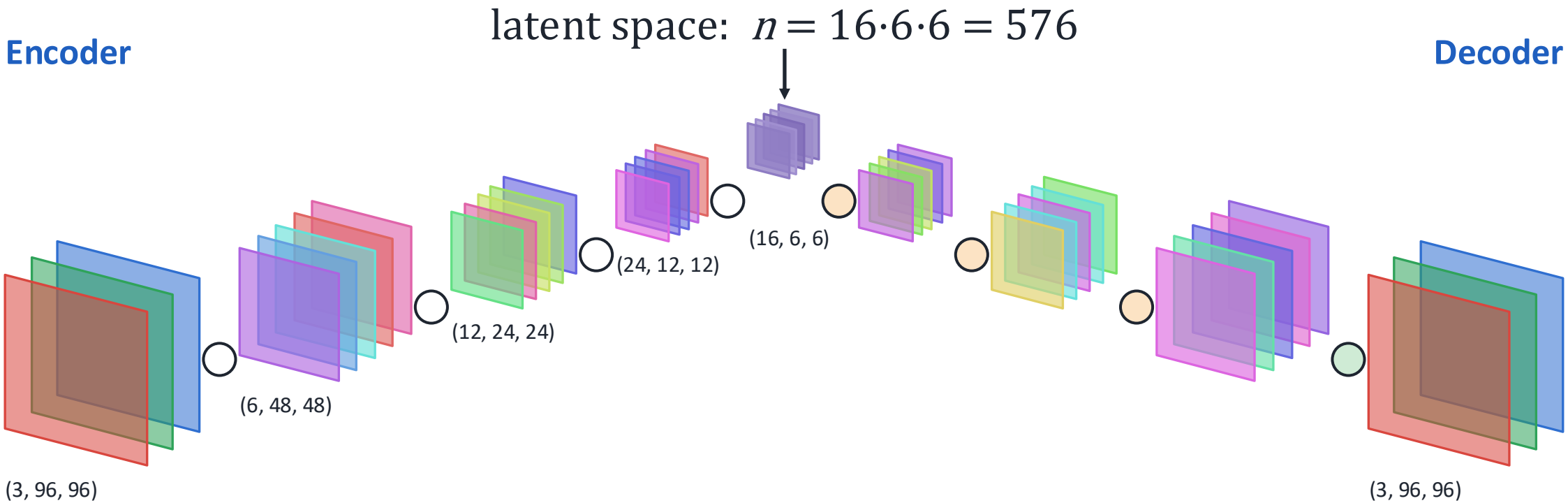


Convolutional Autoencoders (2)

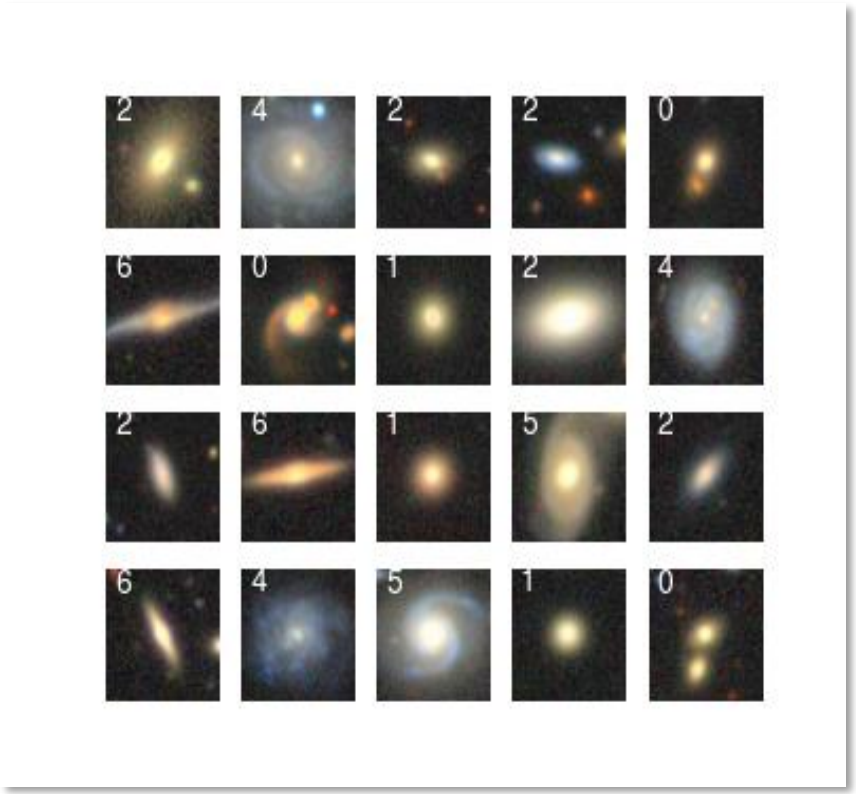
◯ = `maxpool( relu( conv2d(x) ) )`

◯ = `relu( conv_transpose2d(x) )`

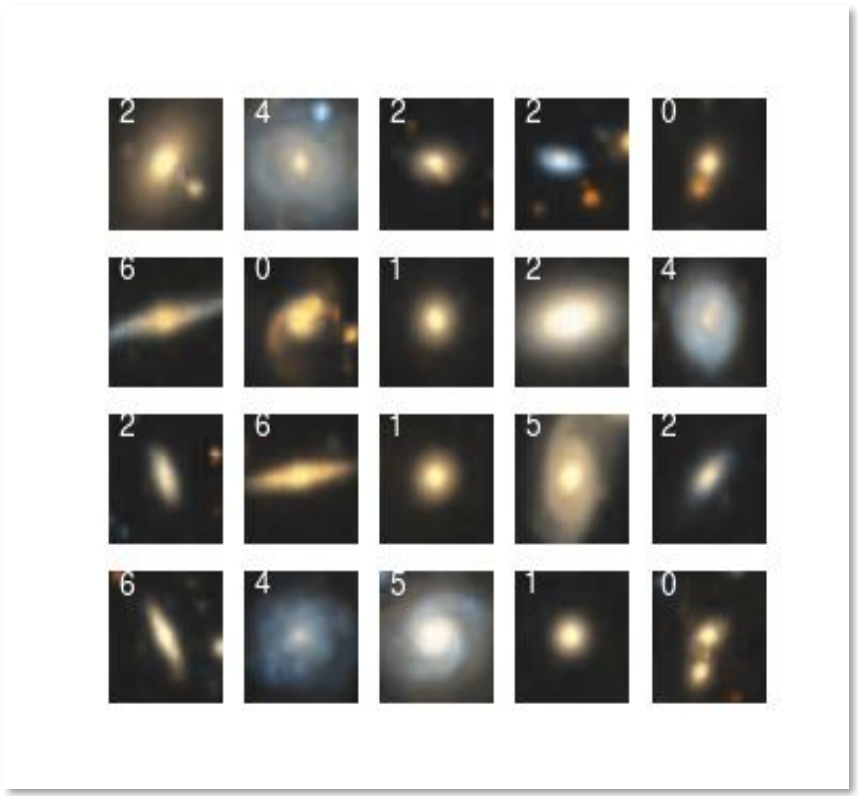
◯ = `sigmoid( conv_transpose2d(x) )`



Convolutional Autoencoders (3)



Original images



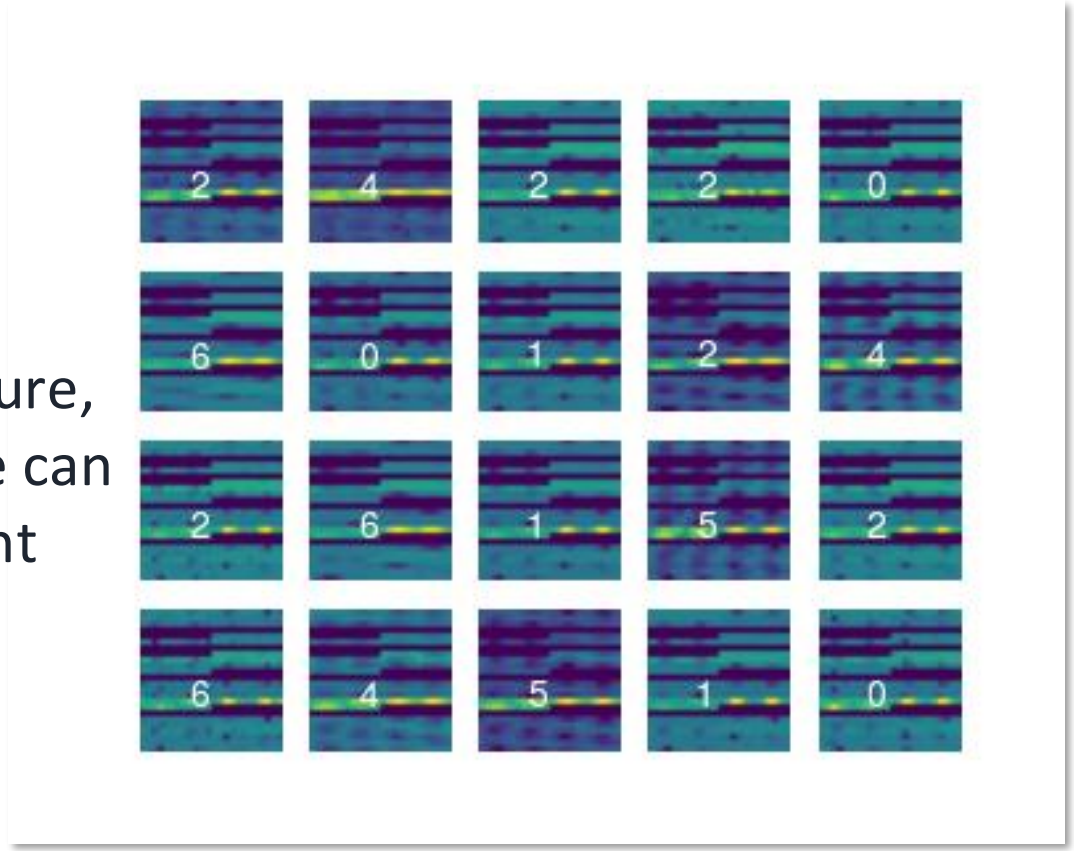
Reconstructed images

tutorial_CAE.ipynb

Convolutional Autoencoders (4)

- The latent space consists of images of shape (16, 6, 6), which we have mapped to single-channel images of shape (1, 24, 24).

- Despite claims to the contrary in the literature, it seems exceedingly unlikely to me that we can ever fully interpret what we see in the latent space!



Latent space images

Summary

- An autoencoder is a model that implements the map $f: x \rightarrow x$ via a bottleneck called the latent space whose dimensionality is (usually) much smaller than that of the input space.

- The applications include:
  1. Data compression
  2. Anomaly detection
  3. Noise removal